

Logic Tensor Networks for Knowledge-Infused Learning: Foundations, Experiments, and Applications

Final Master Thesis

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Índex

1 Introduction

2 LTN

3 Experimental Setting

4 Conclusions

Índex

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2 LTN

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Introduction: Modern ML and Symbolic AI

Modern ML

- Excellent at pattern extraction
- Scales with large datasets
- Strong predictive performance

- Poor OOD generalization
- Data inefficiency
- Lack of interpretability

Symbolic AI

- Human readable symbols
- Based on rich knowledge
- Success at many tasks

- Dealing noisy/inaccurate data
- High computational complexity
- Lack robustness to outliers

Introduction: Neurosymbolic AI and LTN

Neurosymbolic Learning arises as an attempt to combine to combine these paradigms, aiming to achieve models capable of:

- Learn from complex data itself.
- Reason with complex representations of knowledge.
- Provide through rich knowledge a model of the world that complements data.

Introduction: Neurosymbolic AI and LTN

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Logic Tensor Networks (LTN) arise as a new neurosymbolic learning paradigm able to provide a practical instantiation of this idea.

Introduction: Goals

The **goals** of this thesis and presentation are:

- Introduce LTN as a tool
- Provide a guideline for their application.
- Explore and show their potential, usefulness and utilities
- Illustrate them through an example

Índex

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LTN: Description and key concepts

LTN

Logic Tensor Networks (LTN) are a neurosymbolic framework and computational model that supports learning and reasoning about data using a knowledge base composed of a set of first order logic (FOL) axioms as the objective of a neural network model.

LTN: Description and key concepts

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Key concepts:

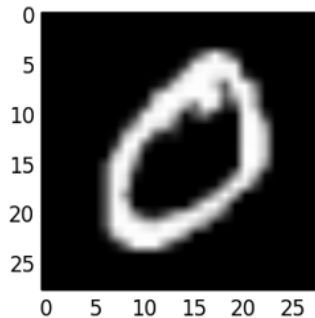
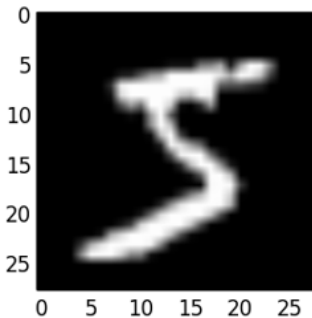
- Real Logic
- Fuzzy Logic
- Grounding
- Logical operators

LTN: Example

To illustrate the theoretical elements of this thesis, we are going to go through an example.

The problem

Predicting the sum of the two natural numbers between 0 and 9 showed on two MNIST images:



LTN: Real Logic

Real Logic

Real Logic is a fully differentiable logical language defined in a first-order language $\mathcal{L}$ with a signature that contains a set $\mathcal{C}$ of constant symbols (objects), a set $\mathcal{F}$ of functional symbols, a set $\mathcal{P}$ of relational symbols (also called predicates) and a set $\mathcal{X}$ of variable symbols.

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Variables and constants can be of different types $\rightarrow$ Need to assign types to the elements of $\mathcal{L}$
Functions:

- $\mathbf{D} : \mathcal{X} \cup \mathcal{C} \rightarrow \mathcal{D}$
- $\mathbf{D}_{in} : \mathcal{F} \cup \mathcal{P} \rightarrow \mathcal{D}^*$
- $\mathbf{D}_{out} : \mathcal{F} \rightarrow \mathcal{D}$

Assign to the constants, variables their domains and to functions and predicates the domains of their inputs and outputs.

LTN: Real Logic Example

Variables

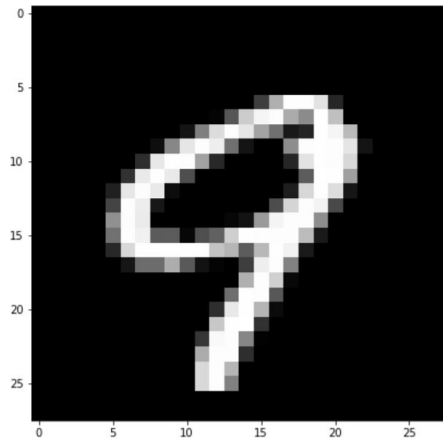
- x, y ranging over the MNIST in the data
- d_1, d_2 ranging over digits
- n for the labels (result of each addition)

Domains

- $\mathbf{D}(x) = \mathbf{D}(y) = \textit{images}$ (MNIST digit images)
- $\mathbf{D}(d_1) = \mathbf{D}(d_2) = \textit{digits}$ (digits from 0 to 9)
- $\mathbf{D}(n) = \textit{results}$ (integers labeling additions results)

Predicates

- $\textit{digit}(x, d)$: probability of image x being digit d



LTN: Fuzzy Logic

We have seen so far:

Predicates

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LTN: Fuzzy Logic

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Predicates

- $digit(x, d)$: probability of image x being digit d

This is caused by the type of logic LTN implement:

Fuzzy Logic

Since in real world scenarios things usually are not 100% true or false, LTN adopt a **Fuzzy Logic** in which measures the degree of truth of statements in the interval $[0, 1]$.

LTN: Grounding

Main problem of Neurosymbolic learning

How do we interpretate symbols in real data?

Grounding

A **Grounding** ($\mathcal{G}$) for a logical language $\mathcal{L}$ on the set of domains $\mathcal{D}$ provides the interpretation of both the domain symbols in $\mathcal{D}$ and the non-logical symbols in $\mathcal{L}$.

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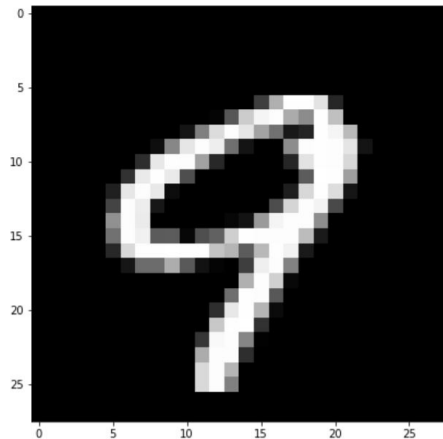
Intuitively, $\mathcal{G}$ associates to:

- Constant c of $\mathcal{L}$: Tensor of real numbers which represent the numeric features of the objects denoted by c .
- Predicate p of $\mathcal{L}$: Real number in $[0, 1]$ representing the degree of confidence in the truth of p .

LTN: Grounding Example

Groundings

- $\mathcal{G}(\text{images}) = [0, 1]^{28 \times 28 \times 1}$.
- $\mathcal{G}(\text{results}) = \mathbb{N}$.
- $\mathcal{G}(\text{digits}) = \{0, 1, \dots, 9\}$
- $\mathcal{G}(x) \in [0, 1]^{m \times 28 \times 28 \times 1}$, $\mathcal{G}(y) \in [0, 1]^{m \times 28 \times 28 \times 1}$,
 $\mathcal{G}(n) \in [0, 1]^m$
- $\mathcal{G}(d_1) = \mathcal{G}(d_2) = \langle 0, 1, \dots, 9 \rangle$
- $\mathcal{G}(\text{digit}|\theta) : x, d \rightarrow \text{onehot}(d)^T \cdot \text{softmax}(\text{CNN}_\theta(x))$



LTN: Logic Operators

To build logic formulas we also need **Logical Operators**.

They may not be stable (i.e. gradient vanishing)

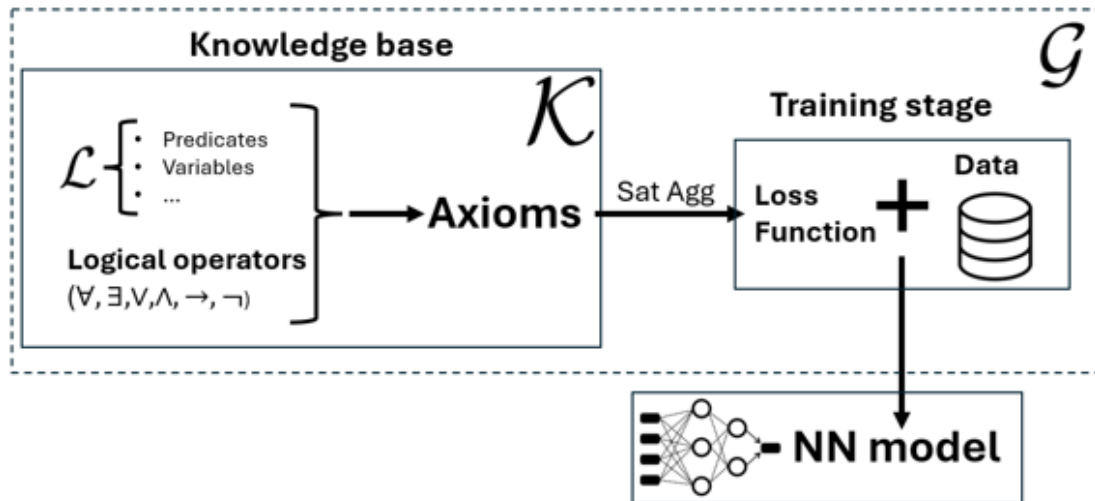
Real product configuration:

- Negation ($\neg$) : $N_S(a) = 1 - a$
- Conjunction ($\wedge$) : $T_P(a, b) = ab$
- Disjunction ($\vee$) : $S_P(a, b) = a + b - ab$
- Implication ($\rightarrow$) : $I_R(a, b) = 1 - a + ab$
- Exists ($\exists$) : $A_{pM}(a_1, \dots, a_n) = \left(\frac{1}{n} \sum_{i=1}^n a_i^p\right)^{\frac{1}{p}} \quad p \geq 1$
- For all ($\forall$) : $A_{pME}(a_1, \dots, a_n) = 1 - \left(\frac{1}{n} \sum_{i=1}^n (1 - a_i)^p\right)^{\frac{1}{p}} \quad p \geq 1$

Combining them with $\mathcal{L}$ we can build **Axioms**.

LTN: Learning and architecture

Grounded



Índex

- 1 Introduction
- 2 LTN
- 3 Experimental Setting**
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Experimental Setting: General Description

Goal: Predicting the weight of people based on their height with a 1-D linear regressor.

Model: Linear Regressor modeled by a learnable MLP function f^* .

Data: 300 point with $x \in [1.4, 2.0]$ generated according to: $y = 0.9 \cdot (100 \cdot x) + 5 \cdot \sin(3 \cdot x)$

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Experiments Performed:

- **Experiment 1:** BaseLine. There is only one axiom. It imposes $f^*(x_i) = y_i$:
$$\forall_{reg} \text{Diag}(x, y)(f^*(x) = y)$$
- **Experiment 2:** Additionally, there is a reasonable axiom inducing locally increasing monotony:
$$\forall_{mon} (x_1, x_2) \in N_3(x) ((x_1 \leq x_2) \Rightarrow (f(x_1) \leq f(x_2)))$$
- **Experiment 3:** Instead, there is a misleading axiom inducing locally decreasing monotony:
$$\forall_{mon} (x_1, x_2) \in N_3(x) ((x_1 \leq x_2) \Rightarrow (f(x_2) \leq f(x_1)))$$

Experimental Setting: Results

-	Experiment 1	Experiment 2	Experiment 3
Train Reg Sat	0.8849	0.9123	0.8848
Test Reg Sat	0.8873	0.9017	0.8790
OOD Regression Sat	0.7466	0.8019	0.7391
Train Mon Sat	-	0.9458	0.5545
Test Mon Sat	-	0.9448	0.5535
OOD Mon Sat	-	0.8632	0.6368
Train RMSE	10.6084	7.8394	10.8009
Test RMSE	10.5173	9.0493	11.0736
OOD RMSE	27.0991	20.0528	28.0119

Índex

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- 2 LTN
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Conclusions

- LTN arise as a successful knowledge transferring mechanism that allows to translate human-readable logical statements into differentiable, optimizable functions.
- Accurate/reasonable hypothesizes upgrade results, whereas misleading ones can worse them.
- LTN provide a whole new level of explainability both on queries as well as on measuring the satisfaction degree of axioms for a certain prediction.

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Thank you so much for your attention!

Questions?

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